

VelOps



Allister Kohn

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Who am I?

- Allister, 23 years old
- Based in the Paris region
- Scientific Baccalaureate (2019)
- Graduate Engineer (2024)
- MSc in Data Sciences & Business Analytics (2025)
- Career focus:
Data Science → ML Engineering → MLOps



Vélib'

- Smovengo is the operator of **Vélib' Métropole**, the public bike-sharing service of the Paris metropolitan area
- Vélib' is the **largest station-based bike-sharing system** in the world, covering Paris and the wider Île-de-France region
- Smovengo provides open access to Vélib's real-time operational data through a **public API**

Key figures:

- 1 station approximately every 300 meters in Paris
- 1,500 stations across Paris and the Île-de-France region
- 20,000 bicycles, including 40% electric-assisted bikes
- 940,000 unique users in 2025

Smovengo

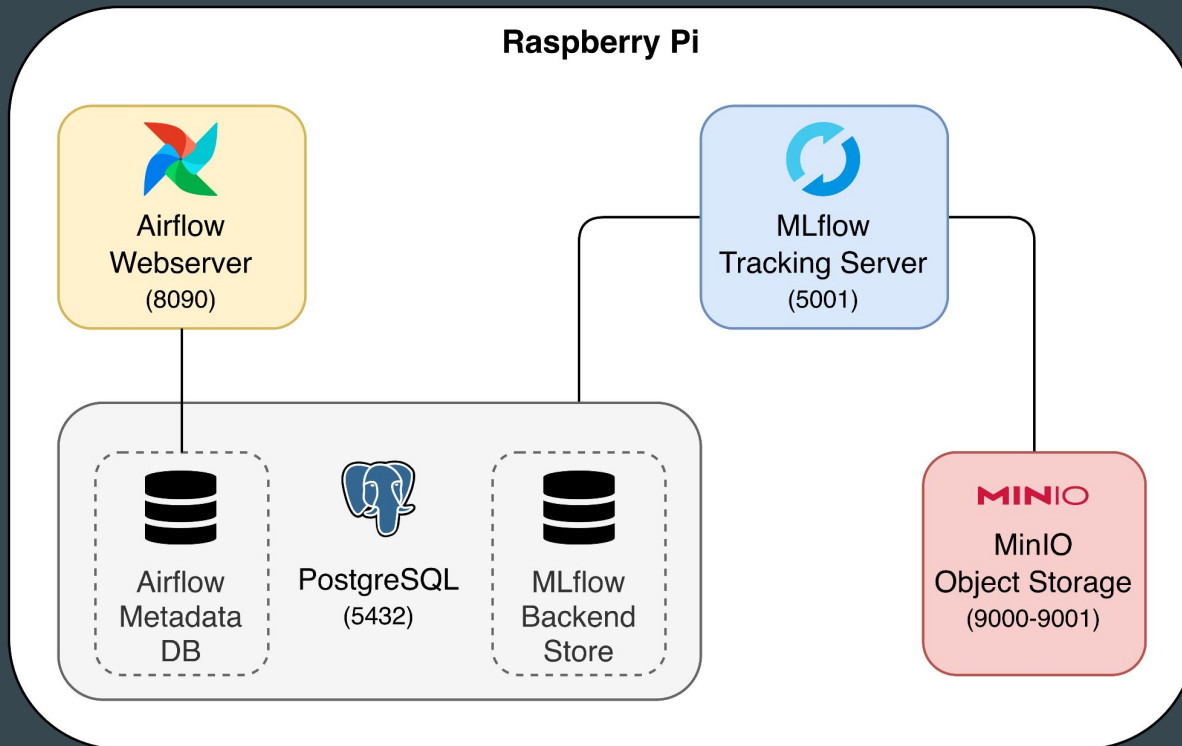


Introduction

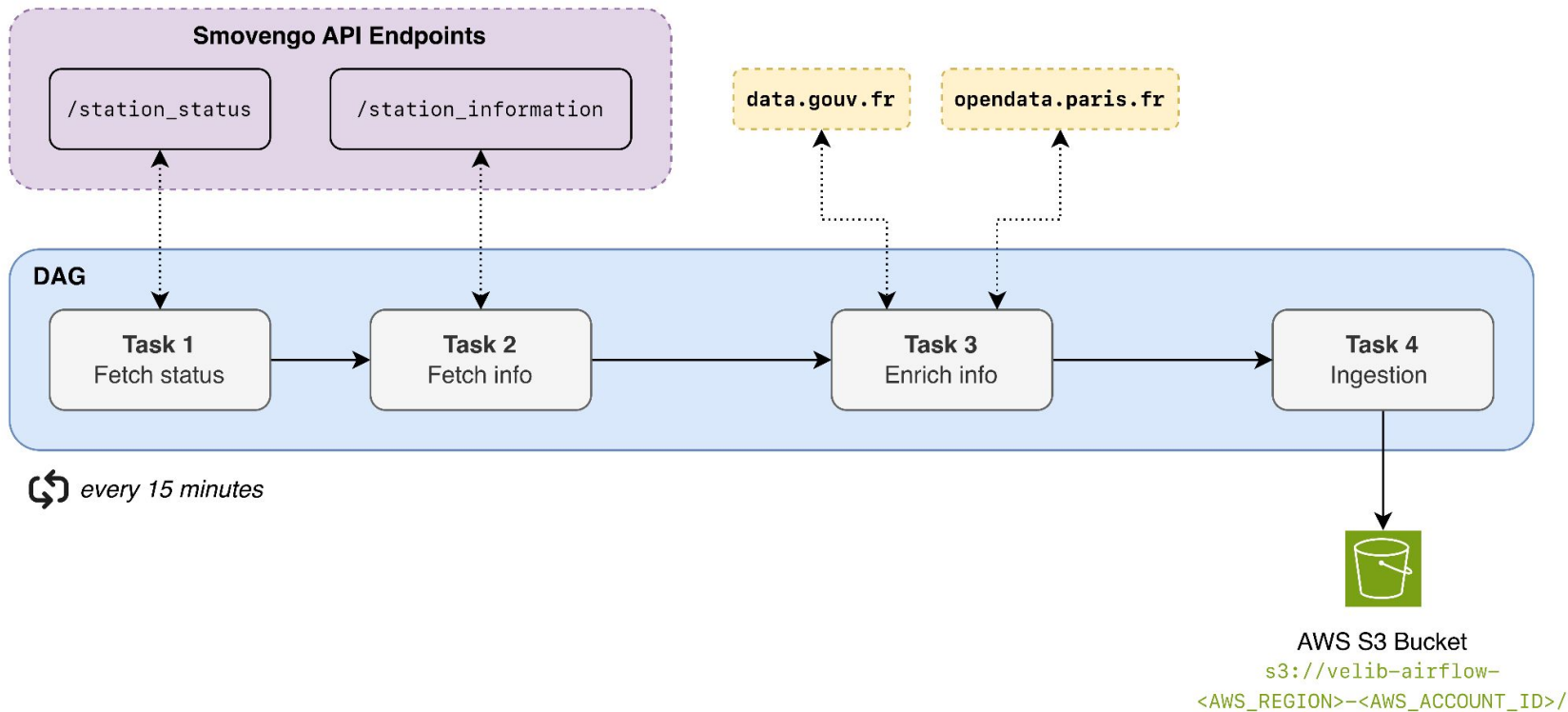
- VelOps is an end-to-end **machine learning project** using Vélib' data
- Its goal is to **predict bike availability** states at station level within a two-hour horizon
- The system runs on a Raspberry Pi, highlighting a lightweight and resource-efficient setup
- The project relies on a broad set of **modern data technologies** across the full workflow
- The project is structured into two independent components: one dedicated to **data ingestion and transformation**, and another focused on **machine learning**



Infrastructure



Data Ingestion



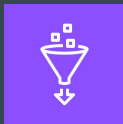


Data Storage: Amazon S3

Amazon S3 acts as the raw storage layer of the pipeline.

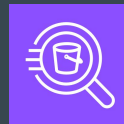
Data is ingested and stored as CSV files using a date-partitioned structure.

```
station_status/raw/  
└─ year/  
    └─ month/  
        └─ day/  
            └─ hour-minute.csv
```



Glue Data Catalog & Athena

Terraform defines the Glue Catalog schema, partitions and S3 locations for the raw CSV files. Glue acts as a metadata layer, while Athena uses that metadata to query S3 data directly with SQL.



Data Transformation: dbt

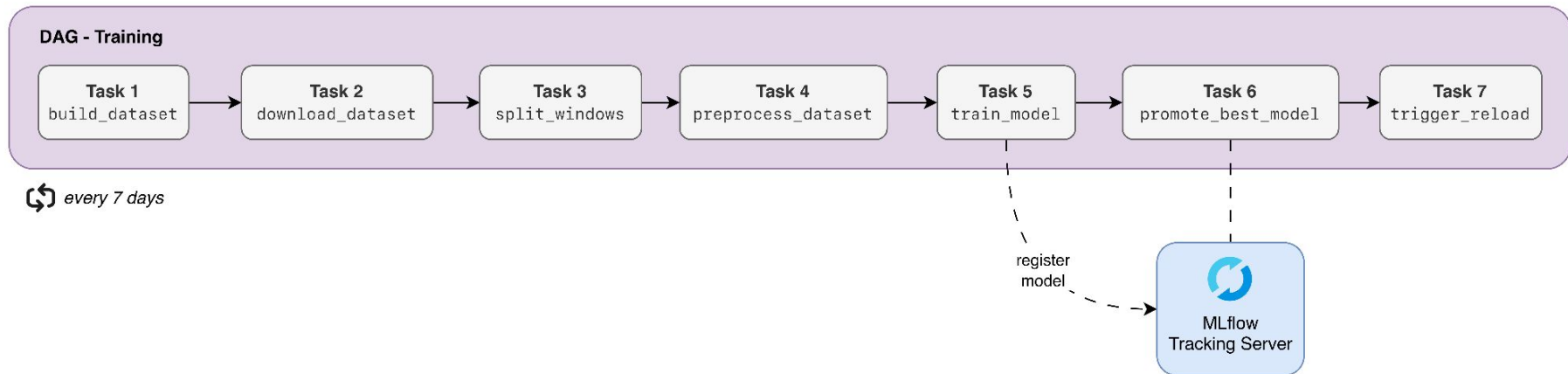
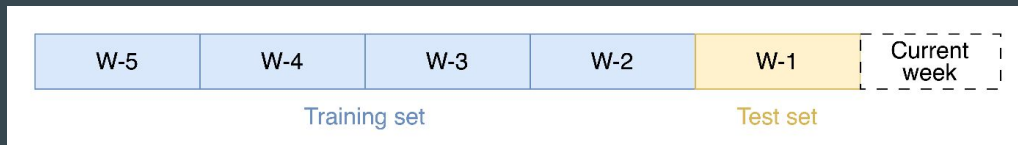
dbt uses Athena as its SQL execution engine to transform raw ingestion tables into machine-learning-ready Parquet datasets.

The pipeline runs incrementally and writes partitioned feature tables back into S3 for downstream machine learning training.



Machine Learning

The dataset covers a **5-week period**, corresponding to 35 days = 840 hours = 3,360 15-minute intervals. Given the 1,500 stations in the network, this results in a total of **5,040,000 status records** (> 700 Mb).



Sources

- <https://www.group-indigo.com/metier/smovengo/>
- <https://www.smovengo.fr/>
- <https://www.velib-metropole.fr/>





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